

Q1 - 24 June - Shift 1

Let $f(\theta) = \sin \theta + \int_{-\pi/2}^{\pi/2} (\sin \theta + t \cos \theta) f(t) dt$. Then the

value of $\left| \int_0^{\pi/2} f(\theta) d\theta \right|$ is _____.

Space for your notes:

Q2 - 24 June - Shift 1

Let $\text{Max}_{0 \leq x \leq 2} \left\{ \frac{9-x^2}{5-x} \right\} = \alpha$ and $\text{Min}_{0 \leq x \leq 2} \left\{ \frac{9-x^2}{5-x} \right\} = \beta$

If $\int_{\beta - \frac{8}{3}}^{2\alpha - 1} \text{Max} \left\{ \frac{9-x^2}{5-x}, x \right\} dx = \alpha_1 + \alpha_2 \log_e \left(\frac{8}{15} \right)$ then

$\alpha_1 + \alpha_2$ is equal to _____

Space for your notes:

Q3 - 24 June - Shift 2

The value of the integral

$\int_{-\pi/2}^{\pi/2} \frac{dx}{(1+e^x)(\sin^6 x + \cos^6 x)}$ is equal to

(A) 2π

(B) 0

(C) π

(D) $\frac{\pi}{2}$

Space for your notes:

Q4 - 24 June - Shift 2

Questions

MathonGo

$$\lim_{n \rightarrow \infty} \left(\frac{n^2}{(n^2+1)(n+1)} + \frac{n^2}{(n^2+4)(n+2)} + \frac{n^2}{(n^2+9)(n+3)} + \dots + \frac{n^2}{(n^2+n^2)(n+n)} \right)$$

is equal to

- (A) $\frac{\pi}{8} + \frac{1}{4} \log_e 2$ (B) $\frac{\pi}{4} + \frac{1}{8} \log_e 2$
 (C) $\frac{\pi}{4} - \frac{1}{8} \log_e 2$ (D) $\frac{\pi}{8} + \log_e \sqrt{2}$

Space for your notes:

Q5 - 25 June - Shift 1

The value of $\int_0^{\pi} \frac{e^{\cos x} \sin x}{(1 + \cos^2 x)(e^{\cos x} + e^{-\cos x})} dx$ is

equal to

- (A) $\frac{\pi^2}{4}$ (B) $\frac{\pi^2}{2}$
 (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

Space for your notes:

Q6 - 25 June - Shift 1

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Questions

MathonGo

Let $g : (0, \infty) \rightarrow \mathbb{R}$ be a differentiable function such that

$$\int \left(\frac{x(\cos x - \sin x)}{e^x + 1} + \frac{g(x)(e^x + 1 - xe^x)}{(e^x + 1)^2} \right) dx = \frac{xg(x)}{e^x + 1} + c,$$

for all $x > 0$, where c is an arbitrary constant.

Then,

(A) g is decreasing in $\left(0, \frac{\pi}{4}\right)$

(B) g' is increasing in $\left(0, \frac{\pi}{4}\right)$

(C) $g + g'$ is increasing in $\left(0, \frac{\pi}{2}\right)$

(D) $g - g'$ is increasing in $\left(0, \frac{\pi}{2}\right)$

Space for your notes:

Q7 - 25 June - Shift 2

If $b_n = \int_0^{\frac{\pi}{2}} \frac{\cos^2 nx}{\sin x} dx$, $n \in \mathbb{N}$, then

(A) $b_3 - b_2, b_4 - b_3, b_5 - b_4$ are in an A.P. with common difference -2

(B) $\frac{1}{b_3 - b_2}, \frac{1}{b_4 - b_3}, \frac{1}{b_5 - b_4}$ are in an A.P. with common difference 2

(C) $b_3 - b_2, b_4 - b_3, b_5 - b_4$ are in a G.P.

(D) $\frac{1}{b_3 - b_2}, \frac{1}{b_4 - b_3}, \frac{1}{b_5 - b_4}$ are in an A.P. with common difference -2

Space for your notes:

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Q8 - 25 June - Shift 2

The value of $b > 3$ for which

$$12 \int_3^b \frac{1}{(x^2 - 1)(x^2 - 4)} dx = \log_e \left(\frac{49}{40} \right), \text{ is equal to}$$

Space for your notes:

Q9 - 26 June - Shift 1

Let $f(x) = \max\{|x + 1|, |x + 2|, \dots, |x + 5|\}$. Then

$$\int_{-6}^0 f(x) dx \text{ is equal to } \underline{\hspace{2cm}}.$$

Space for your notes:

Q10 - 26 June - Shift 1

The value of the integral

$$\frac{48}{\pi^4} \int_0^{\pi} \left(\frac{3\pi x^2}{2} - x^3 \right) \frac{\sin x}{1 + \cos^2 x} dx \text{ is equal to}$$

_____.

Space for your notes:

Q11 - 26 June - Shift 2

$$\text{The integral } \frac{24}{\pi} \int_0^{\sqrt{2}} \frac{(2 - x^2) dx}{(2 + x^2)\sqrt{4 + x^4}} \text{ is equal to } \underline{\hspace{2cm}}.$$

Space for your notes:

Q12 - 27 June - Shift 1

$$\text{The value of the integral } \int_{-2}^2 \frac{|x^3 + x|}{(e^{x|x|} + 1)} dx \text{ is equal}$$

Space for your notes:

to :

(A) $5e^2$

(B) $3e^{-2}$

(C) 4

(D) 6

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Q13 - 27 June - Shift 2

Let f be a differentiable function in $\left(0, \frac{\pi}{2}\right)$.

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If $\int_{\cos x}^1 t^2 f(t) dt = \sin^3 x + \cos x$ then $\frac{1}{\sqrt{3}} f'\left(\frac{1}{\sqrt{3}}\right)$ is

equal to :

(A) $6 - 9\sqrt{2}$ (B) $6 - \frac{9}{\sqrt{2}}$

(C) $\frac{9}{2} - 6\sqrt{2}$ (D) $\frac{9}{\sqrt{2}} - 6$

Q14 - 27 June - Shift 2

The integral $\int_0^1 \frac{1}{7^{\left[\frac{1}{x}\right]}} dx$, where $[.]$ denotes the

Space for your notes:

greatest integer function is equal to

(A) $1 + 6 \log_e \left(\frac{6}{7}\right)$ (B) $1 - 6 \log_e \left(\frac{6}{7}\right)$

(C) $\log_e \left(\frac{7}{6}\right)$ (D) $1 - 7 \log_e \left(\frac{6}{7}\right)$

Q15 - 28 June - Shift 1

Let $[t]$ denote the greatest integer less than or equal to t . Then, the value of the integral

Space for your notes:

$\int_0^1 [-8x^2 + 6x - 1] dx$ is equal to

(A) -1 (B) $-\frac{5}{4}$

(C) $\frac{\sqrt{17} - 13}{8}$ (D) $\frac{\sqrt{17} - 16}{8}$

Q16 - 28 June - Shift 2

Let $f : \mathbf{R} \rightarrow \mathbf{R}$ be a differentiable function such that $f\left(\frac{\pi}{4}\right) = \sqrt{2}$, $f\left(\frac{\pi}{2}\right) = 0$ and $f'\left(\frac{\pi}{2}\right) = 1$ and

let $g(x) = \int_x^{\pi/4} (f'(t) \sec t + \tan t \sec t f(t)) dt$ for

$x \in \left[\frac{\pi}{4}, \frac{\pi}{2}\right)$. Then $\lim_{x \rightarrow \left(\frac{\pi}{2}\right)^-} g(x)$ is equal to

- (A) 2 (B) 3
(C) 4 (D) -3

Space for your notes:

Q17 - 28 June - Shift 2

Let $f : \mathbf{R} \rightarrow \mathbf{R}$ be continuous function satisfying $f(x) + f(x+k) = n$, for all $x \in \mathbf{R}$ where $k > 0$ and n

is a positive integer. If $I_1 = \int_0^{4nk} f(x) dx$ and

$I_2 = \int_{-k}^{3k} f(x) dx$, then

- (A) $I_1 + 2I_2 = 4nk$ (B) $I_1 + 2I_2 = 2nk$
(C) $I_1 + nI_2 = 4n^2k$ (D) $I_1 + nI_2 = 6n^2k$

Space for your notes:

Q18 - 29 June - Shift 1

Questions

MathonGo

$$\int_0^5 \cos \left(\pi \left(x - \left[\frac{x}{2} \right] \right) \right) dx,$$

Space for your notes:

Where $[t]$ denotes greatest integer less than or equal to t , is equal to :

- (A) -3 (B) -2
(C) 2 (D) 0

Q19 - 29 June - Shift 2

Let f be a real valued continuous function on $[0,1]$

Space for your notes:

and $f(x) = x + \int_0^1 (x-t)f(t)dt$. Then which of the

following points (x,y) lies on the curve $y = f(x)$?

- (A) $(2, 4)$ (B) $(1, 2)$
(C) $(4, 17)$ (D) $(6, 8)$

Q20 - 29 June - Shift 2

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$$\text{If } \int_0^2 \left(\sqrt{2x} - \sqrt{2x - x^2} \right) dx =$$

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$$\int_0^1 \left(1 - \sqrt{1 - y^2} - \frac{y^2}{2} \right) dy + \int_1^2 \left(2 - \frac{y^2}{2} \right) dy + I$$

$$(A) \int_0^1 \left(1 + \sqrt{1 - y^2} \right) dy$$

$$(B) \int_0^1 \left(\frac{y^2}{2} - \sqrt{1 - y^2} + 1 \right) dy$$

$$(C) \int_0^1 \left(1 - \sqrt{1 - y^2} \right) dy$$

$$(D) \int_0^1 \left(\frac{y^2}{2} + \sqrt{1 - y^2} + 1 \right) dy$$

Answer Key

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Q1 (1) Q2 (34) Q3 (C) Q4 (A)
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Q5 (C) Q6 (D) Q7 (D) Q8 (6)
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Q9 (21) Q10 (6) Q11 (3) Q12 (D)
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Q13 (B) Q14 (A) Q15 (C) Q16 (B)
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Q17 (C) Q18 (D) Q19 (D) Q20 (C)
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Q1 (1)

$$f(\theta) = \sin \theta + \int_{-\pi/2}^{\pi/2} (\sin \theta + t \cos \theta) f(t) dt$$

$$f(\theta) = \sin \theta + \sin \theta \int_{-\pi/2}^{\pi/2} f(t) dt + \cos \theta \int_{-\pi/2}^{\pi/2} t f(t) dt$$

$$\text{Let } A = \int_{-\pi/2}^{\pi/2} f(t) dt, B = \int_{-\pi/2}^{\pi/2} t f(t) dt$$

$$f(\theta) = \sin \theta + A \sin \theta + B \cos \theta$$

$$f(\theta) = (A + 1) \sin \theta + B \cos \theta$$

$$A = \int_{-\pi/2}^{\pi/2} (A + 1) \sin t + B \cos t dt$$

$$A = 2B \dots\dots(1)$$

$$B = \int_{-\pi/2}^{\pi/2} t((A + 1) \sin t + B \cos t) dt$$

$$B = \int_{-\pi/2}^{\pi/2} t(A + 1) \sin t dt$$

$$B = (A + 1) 2 \int_0^{\pi/2} t \sin t dt$$

$$B = (A + 1) 2.1$$

$$2A + 2 - B = 0 \dots\dots(2)$$

After solving

$$B = -\frac{2}{3}, A = -\frac{4}{3}$$

$$\left| \int_0^{\pi/2} f(\theta) d\theta \right| = \left| \int_0^{\pi/2} -\frac{1}{3} \sin \theta - \frac{2}{3} \cos \theta \right|$$

$$= 1$$

Q2 (34)

$$y = \frac{9-x^2}{5-x} = 5 + x + \frac{16}{x-5}$$

$$\frac{dy}{dx} = 1 - \frac{16}{(x-5)^2}$$

So critical point is $x = 1$ in $[0, 2]$

$$y(0) = \frac{9}{5}, \quad y(1) = 2, \quad y(2) = \frac{5}{3}$$

$$\text{So } \alpha = 2 \text{ and } \beta = \frac{5}{3}$$

$$I = \int_{-1}^3 \max\left(\frac{9-x^2}{5-x}, x\right) dx$$

$$I = \int_{-1}^{9/5} \frac{9-x^2}{5-x} dx + \int_{9/5}^3 x dx$$

$$I = \int_{-1}^{9/5} 5 + x + \frac{16}{x-5} dx + \int_{9/5}^3 x dx$$

After solving

$$I = 14 + \frac{28}{25} + 16 \ln\left(\frac{8}{15}\right) + \frac{72}{25}$$

$$\alpha_1 = 18 \text{ and } \alpha_2 = 16$$

Q3 (C)

$$I = \int_{-\pi/2}^0 \frac{dx}{(1+e^x)(\sin^6 x + \cos^6 x)} + \int_0^{\pi/2} \frac{dx}{(1+e^x)(\sin^6 x + \cos^6 x)}$$

Put $x = -t$

$$= \int_{\pi/2}^0 \frac{-dt}{(1+e^{-t})(\sin^6 t + \cos^6 t)} + \int_0^{\pi/2} \frac{dx}{(1+e^x)(\sin^6 x + \cos^6 x)}$$

$$= \int_0^{\pi/2} \frac{(e^x + 1)dx}{(1+e^x)(\sin^6 x + \cos^6 x)}$$

$$= \int_0^{\pi/2} \frac{dx}{(\sin^2 x + \cos^2 x)(\sin^4 x - \sin^2 x \cos^2 x + \cos^4 x)}$$

$$= \int_0^{\pi/2} \frac{(1 + \tan^2 x) \sec^2 x dx}{(\tan^4 x - \tan^2 x + 1)}$$

Put $\tan x = t$

$$= \int_0^{\infty} \frac{(1+t^2)dt}{(t^4 - t^2 + 1)}$$

$$= \int_0^{\infty} \frac{\left(1 + \frac{1}{t^2}\right)dt}{t^2 - 1 + \frac{1}{t^2}} = \int_0^{\infty} \frac{\left(1 + \frac{1}{t^2}\right)dt}{\left(t - \frac{1}{t}\right)^2 + 1}$$

Put $t - \frac{1}{t} = z$

$$\left(1 + \frac{1}{t^2}\right)dt = dz$$

$$= \int_{-\infty}^{\infty} \frac{dz}{1+z^2} = \left(\tan^{-1} z\right)_{-\infty}^{\infty}$$

$$= \frac{\pi}{2} - \left(-\frac{\pi}{2}\right) = \pi$$

Q4 (A)

$$\begin{aligned} & \lim_{n \rightarrow \infty} \left(\sum_{r=1}^n \frac{n^2}{(n^2 + r^2)(n+r)} \right) \\ &= \lim_{n \rightarrow \infty} \left(\sum_{r=1}^n \frac{1}{n \left(1 + \left(\frac{r}{n} \right)^2 \right) \left(1 + \left(\frac{r}{n} \right) \right)} \right) \\ &= \int_0^1 \frac{dx}{(1+x^2)(1+x)} = \frac{1}{2} \int_0^1 \frac{1-x}{1+x^2} dx + \frac{1}{2} \int_0^1 \frac{1}{1+x} dx \\ &= \frac{1}{2} \int_0^1 \left(\frac{1}{1+x^2} - \frac{x}{1+x^2} \right) dx + \frac{1}{2} (\ln(1+x))_0^1 \\ &= \frac{1}{2} \left[\tan^{-1} x - \frac{1}{2} \ln(1+x^2) \right]_0^1 + \frac{1}{2} \ln 2 \\ &= \frac{1}{2} \left[\frac{\pi}{4} - \frac{1}{2} \ln 2 \right] + \frac{1}{2} \ln 2 \\ &= \frac{\pi}{8} + \frac{1}{4} \ln 2 \end{aligned}$$

Q5 (C)

$$\int_0^{\pi} \frac{e^{\cos x} \sin x}{(1 + \cos^2 x)(e^{\cos x} + e^{-\cos x})} dx \dots (1)$$

Use King's property

$$I = \int_0^{\pi} \frac{e^{-\cos x} \sin x}{(1 + \cos^2 x)(e^{-\cos x} + e^{\cos x})} dx \dots (2)$$

On adding equation (1) and (2), we get

$$2I = \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx = 2 \int_0^{\pi/2} \frac{\sin x}{1 + \cos^2 x} dx$$

On putting $\cos x = t$, we get

$$I = \int_0^1 \frac{dt}{1+t^2} = (\tan^{-1} t)_0^1 = \frac{\pi}{4}$$

Q6 (D)

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Hints and Solutions

MathonGo

$$\int \left(\frac{x(\cos x - \sin x)}{e^x + 1} + \frac{g(x)(e^x + 1 - xe^x)}{(e^x + 1)^2} \right) dx = \frac{xg(x)}{e^x + 1} + c$$

On differentiating both sides w.r.t. x , we get

$$\left(\frac{x(\cos x - \sin x)}{e^x + 1} + \frac{g(x)(e^x + 1 - xe^x)}{(e^x + 1)^2} \right)'$$

$$= \frac{(e^x + 1)(g(x) + xg'(x)) - e^x \cdot x \cdot g(x)}{(e^x + 1)^2}$$

$$(e^x + 1)x(\cos x - \sin x) + g(x)(e^x + 1 - xe^x)$$

$$= (e^x + 1)(g(x) + xg'(x)) - e^x \cdot x \cdot g(x)$$

$$\Rightarrow g'(x) = \cos x - \sin x$$

$$\Rightarrow g(x) = \sin x + \cos x + C$$

$g(x)$ is increasing in $(0, \pi/4)$

$$g''(x) = -\sin x - \cos x < 0$$

$\Rightarrow g'(x)$ is decreasing function

$$\text{let } h(x) = g(x) + g'(x) = 2 \cos x + C$$

$$\Rightarrow h'(x) = g'(x) + g''(x) = -2 \sin x < 0$$

$\Rightarrow h$ is decreasing

$$\text{let } \phi(x) = g(x) - g'(x) = 2 \sin x + C$$

$$\Rightarrow \phi'(x) = g'(x) - g''(x) = 2 \cos x > 0$$

$\Rightarrow \phi$ is increasing

Hence option D is correct.

Q7 (D)

$$b_n = \int_0^{\pi/2} \frac{1 + \cos 2nx}{\sin x} dx$$

$$b_{n+1} - b_n = \int_0^{\pi/2} \frac{\cos^2(n+1)x - \cos^2 nx}{\sin x} dx$$

$$= \int_0^{\pi/2} \frac{-\sin(2n+1)x \sin x}{\sin x} dx$$

$$= \left(\frac{\cos(2n+1)x}{2n+1} \right)_0^{\pi/2} = \frac{-1}{2n+1}$$

$\frac{1}{b_3 - b_2}, \frac{1}{b_4 - b_3}, \frac{1}{b_5 - b_4}$ are in A.P. with c.d. = -2

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Q8 (6)

$$\frac{12}{3} \left[\int_3^b \left(\frac{1}{x^2 - 4} - \frac{1}{x^2 - 1} \right) dx \right] = \log \frac{49}{40}$$

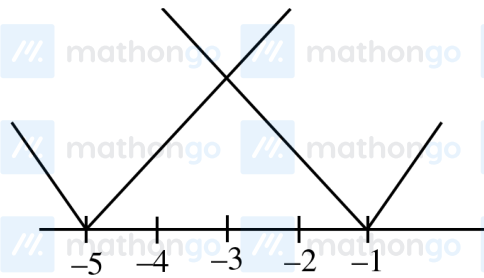
$$\frac{12}{3} \left[\frac{1}{4} \ln \left| \frac{x-2}{x+2} \right| - \frac{1}{2} \ln \left| \frac{x-1}{x+1} \right| \right]_3^b = \log \frac{49}{40}$$

$$\ln \frac{(b-2)(b+1)^2}{(b+2)(b-1)^2} = \ln \frac{49}{50}$$

$$b = 6$$

Q9 (21)

$$f(x) = \max \{ |x+1|, |x+2|, |x+3|, |x+4|, |x+5| \}$$



$$\int_{-6}^0 f(x) dx = \int_{-6}^{-3} |x+1| dx + \int_{-3}^0 |x+5| dx$$

$$= - \int_{-6}^{-3} (x+1) dx + \int_{-3}^0 (x+5) dx$$

$$= - \left[\frac{x^2}{2} + x \right]_{-6}^{-3} + \left[\frac{x^2}{2} + 5x \right]_{-3}^0$$

$$= - \left[\left(\frac{9}{2} - 3 \right) - (18 - 6) \right] + \left[0 - \left(\frac{9}{2} - 15 \right) \right]$$

$$= - \left[\frac{3}{2} - 12 \right] + \frac{21}{2} = \frac{21}{2} + \frac{21}{2} = 21$$

Q10 (6)

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$$I = \frac{48}{\pi^4} \int_0^{\pi} x^2 \left(\frac{3\pi}{2} - x \right) \frac{\sin x}{1 + \cos^2 x} dx \dots (1)$$

Apply king property

$$I = \frac{48}{\pi^4} \int_0^{\pi} (\pi - x)^2 \left(\frac{\pi}{2} + x \right) \frac{\sin x}{1 + \cos^2 x} dx \dots (2)$$

(1) + (2)

$$I = \frac{12}{\pi^3} \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} \left[\pi^2 + (\pi - 2)x(\pi - 2x) \right] dx \dots (3)$$

Apply king again

$$I = \frac{12}{\pi^3} \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} \left[\pi^2 + (\pi - 2)(\pi - x)(2x - \pi) \right] dx \dots (4)$$

(3) + (4)

$$I = \frac{6}{\pi^2} \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} \left[2\pi + (\pi - 2)(\pi - 2x) \right] dx \dots (5)$$

Apply king

$$I = \frac{6}{\pi^2} \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} \left[2\pi + (\pi - 2)(2x - \pi) \right] dx \dots (6)$$

(5) + (6)

$$I = \frac{12}{\pi} \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx$$

Let $\cos x = t \Rightarrow \sin x dx = -dt$

$$I = \frac{12}{\pi} \int_1^{-1} \frac{-dt}{1 + t^2} = 6$$

Q11 (3)

$$\frac{24}{\pi} \int_0^{\sqrt{2}} \frac{(2-x^2)}{(x^2+2)\sqrt{4+x^4}} dx$$

$$\frac{24}{\pi} \int_0^{\sqrt{2}} \frac{x^2 \left(\frac{2}{x^2} - 1 \right) dx}{x \left(x + \frac{2}{x} \right) \times x \sqrt{\frac{4}{x^2} + x^2}}$$

$$\frac{24}{\pi} \int_0^{\sqrt{2}} \frac{\left(\frac{2}{x^2} - 1 \right) dx}{\left(x + \frac{2}{x} \right) \sqrt{\left(x + \frac{2}{x} \right)^2 - 4}}$$

$$x + \frac{2}{x} = t$$

$$dt = \left(1 - \frac{2}{x^2} \right) dx$$

$$I = -\frac{24}{\pi} \int \frac{dt}{t\sqrt{t^2-4}}$$

$$= -\frac{24}{\pi} \times \frac{1}{2} \sec^{-1} \left(\frac{x + \frac{2}{x}}{2} \right) \Bigg|_0^{\sqrt{2}}$$

$$= -\frac{12}{\pi} \left[\sec^{-1} \left(\frac{2\sqrt{2}}{2} \right) - \sec^{-1}(\infty) \right]$$

$$= -\frac{12}{\pi} \left[\frac{\pi}{4} - \frac{2\pi}{2 \times 2} \right] = -\frac{12}{\pi} \left[-\frac{\pi}{4} \right]$$

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$$f(x) = \frac{|x^3 + x|}{(e^{x|x|} + 1)} dx$$

$$\int_{-2}^2 f(x) dx = \int_0^2 (f(x) + f(-x)) dx$$

$$= \int_0^2 \left(\frac{|x^3 + x|}{(e^{x|x|} + 1)} + \frac{|-x^3 - x|}{(e^{-x|-x|} + 1)} \right) dx$$

$$= \int_0^2 \left(\frac{|x^3 + x|}{(e^{x|x|} + 1)} + \frac{|x^3 + x|}{(e^{x|x|} + 1)} \right) dx$$

$$= \int_0^2 \left(\frac{x^3 + x}{(e^{x^2} + 1)} + \frac{x^3 + x}{(e^{x^2} + 1)} \right) dx$$

$$I = \int_0^2 \left(\frac{x^3 + x}{1 + e^{x^2}} + \frac{e^{x^2}(x^3 + x)}{1 + e^{x^2}} \right) dx$$

$$= \int_0^2 (x^3 + x) dx$$

$$= \left[\frac{x^4}{4} + \frac{x^2}{2} \right]_0^2$$

$$= 4 + 2 = 6$$

Q13 (B)

Hints and Solutions

MathonGo

At right hand vicinity of $x = 0$ given equation does not satisfy

$$\therefore \text{LHS} = \int_{-1}^1 t^2 f(t) dt = 0, \text{ RHS} = \lim_{x \rightarrow 0^+} (\sin^3 x + \cos x) = 1$$

LHS $\neq$ RHS hence data given in question is wrong hence BONUS

Correct data should have been

$$\int_{\cos x}^1 t^2 f(t) dt = \sin^3 x + \cos x - 1$$

Calculation for option

differentiating both sides

$$-\cos^2 x f(\cos x) \cdot (-\sin x) = 3\sin^2 x \cdot \cos x - \sin x$$

$$\Rightarrow f(\cos x) = 3 \tan x - \sec^2 x$$

$$\Rightarrow f'(\cos x)(-\sin x) = 3\sec^2 x - 2\sec^2 x \tan x$$

$$\Rightarrow f'(\cos x) \cos x = \frac{2}{\cos^2 x} - \frac{3}{\sin x \cdot \cos x}$$

$$\text{When } \cos x = \frac{1}{\sqrt{3}}; \sin x = \frac{\sqrt{2}}{\sqrt{3}}$$

$$\therefore f'\left(\frac{1}{\sqrt{3}}\right) \frac{1}{\sqrt{3}} = 6 - \frac{9}{\sqrt{2}}$$

Q14 (A)

$$\int_0^1 \frac{1}{7^{\left[\frac{1}{x}\right]}} dx = - \int_1^0 \frac{1}{7^{\left[\frac{1}{x}\right]}} dx$$

$$= (-1) \left[\int_1^{1/2} \frac{1}{7} dx + \int_{1/2}^{1/3} \frac{1}{7^2} dx + \int_{1/3}^{1/4} \frac{1}{7^3} dx + \dots \infty \right]$$

$$= \left(\frac{1}{7} + \frac{1}{2 \cdot 7^2} + \frac{1}{3 \cdot 7^3} + \dots \infty \right) - \left(\frac{1}{7 \cdot 2} + \frac{1}{7^2 \cdot 3} + \frac{1}{7^3 \cdot 4} + \dots \infty \right)$$

$$= -\ln \left(1 - \frac{1}{7} \right) - 7 \left(\frac{1}{7^2 \cdot 2} + \frac{1}{7^3 \cdot 3} + \frac{1}{7^4 \cdot 4} + \dots \infty \right)$$

$$\left[\text{as } \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \infty \right]$$

$$\left[\text{as } \ln(1-x) = - \left(x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots \infty \right) \right]$$

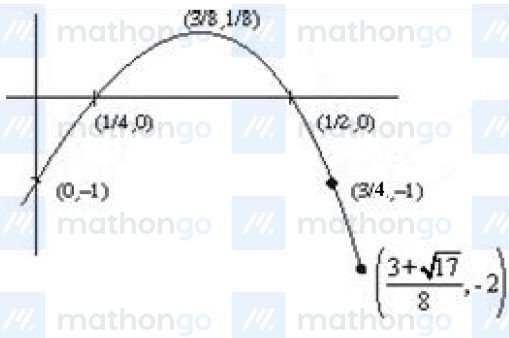
$$= -\ln \frac{6}{7} - 7 \left(-\ln \left(1 - \frac{1}{7} \right) - \frac{1}{7} \right)$$

$$= 6 \ln \frac{6}{7} + 1$$

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Q15 (C)

$$\int_0^1 [-8x^2 + 6x + 1] dx$$

$$= \int_0^{1/4} -1 dx + \int_{1/4}^{1/2} 0 dx + \int_{1/2}^{3/4} -1 dx$$


$$+ \int_{3/4}^{\frac{3+\sqrt{17}}{8}} -2 dx + \int_{\frac{3+\sqrt{17}}{8}}^1 -3 dx$$

$$= -[x]_0^{1/4} + 0 - [x]_{1/2}^{3/4} - 2[x]_{3/4}^{\frac{3+\sqrt{17}}{8}} - 3[x]_{\frac{3+\sqrt{17}}{8}}^1$$

$$= -\left(\frac{1}{4} - 0\right) - \left(\frac{3}{4} - \frac{1}{2}\right) - 2\left(\frac{3+\sqrt{17}}{8} - \frac{3}{4}\right) - 3\left(1 - \frac{3+\sqrt{17}}{8}\right)$$

$$= -\frac{1}{4} - \frac{1}{4} + \frac{-6 - 2\sqrt{17}}{8} + \frac{3}{2} - 3 + \frac{9 + 3\sqrt{17}}{8}$$

$$= \frac{\sqrt{17} - 13}{8}$$

Q16 (B)

$$g(x) = \int_x^{\pi/4} (f'(t) \sec t + \tan t \sec t f(t)) dt$$

$$g(x) = \int_x^{\pi/4} d(f(t) \cdot \sec t) = f(t) \sec t \Big|_x^{\pi/4}$$

$$g(x) = f\left(\frac{\pi}{4}\right) \sec \frac{\pi}{4} - f(x) \cdot \sec x$$

$$g(x) = 2 - f(x) \sec x = 2 - \left(\frac{f(x)}{\cos x} \right)$$

$$\lim_{x \rightarrow \left(\frac{\pi}{2}\right)^-} g(x) = 2 - \lim_{x \rightarrow \left(\frac{\pi}{2}\right)^-} \left(\frac{f(x)}{\cos x} \right)$$

Using L'Hopital Rule

$$= 2 - \lim_{x \rightarrow \left(\frac{\pi}{2}\right)^-} \frac{f'(x)}{(-\sin x)}$$

$$= 2 + \frac{f'\left(\frac{\pi}{2}\right)}{\sin \frac{\pi}{2}} = 2 + \frac{1}{1} = 3$$

Q17 (C)

$$f(x) + f(x+k) = n$$

$$\Rightarrow f(x) = f(x+2k)$$

$f(x)$ is periodic with period $2k$

$$I_1 = \int_0^{4nk} f(x) dx = 2n \int_0^{2k} f(x) dx$$

$$I_2 = \int_{-k}^{3k} f(x) dx = 2 \int_0^{2k} f(x) dx$$

Now,

$$f(x) + f(x+k) = n$$

$$\Rightarrow \int_0^k f(x) dx + \int_0^k f(x+k) dx = nk$$

$$\Rightarrow \int_0^k f(x) dx + \int_k^{2k} f(x) dx = nk$$

$$\Rightarrow \int_0^{2k} f(x) dx = nk$$

$$\Rightarrow I_1 = 2n^2k, I_2 = 2nk$$

$$\Rightarrow I_1 + nI_2 = 4n^2k$$

Q18 (D)

$$I = \int_0^5 \cos\left(\pi x - \pi \left\lfloor \frac{x}{2} \right\rfloor\right) dx$$

$$\Rightarrow I = \int_0^2 \cos(\pi x) dx + \int_2^4 \cos(\pi x - \pi) dx + \int_4^5 \cos(\pi x - 2\pi) dx$$

$$\Rightarrow I = \left[\frac{\sin \pi x}{\pi} \right]_0^2 + \left[\frac{\sin(\pi x - \pi)}{\pi} \right]_2^4 + \left[\frac{\sin(\pi x - 2\pi)}{\pi} \right]_4^5$$

$$\Rightarrow I = 0$$

Q19 (D)

$$f(x) = \left(1 + \int_0^1 f(t) dt \right) x - \int_0^1 t f(t) dt$$

$$f(x) = Ax - B \quad \dots(i)$$

$$A = 1 + \int_0^1 f(t) dt = 1 + \int_0^1 (At - B) dt$$

$$\Rightarrow A = 2(1 - B) \quad \dots(ii)$$

$$\text{Also } B = \int_0^1 t f(t) dt = \int_0^1 (At^2 - Bt) dt$$

$$A = \frac{9}{2} B \quad \dots(iii)$$

From (2), (3)

$$A = \frac{18}{13}, B = \frac{4}{13}$$

$$\text{so } f(6) = 8$$

Q20 (C)

$$\text{LHS} = \int_0^2 \left(\sqrt{2x} - \sqrt{2x - x^2} \right) dx = \frac{8}{3} - \frac{\pi}{2}$$

$$\text{RHS} = \int_0^1 \left(1 - \sqrt{1 - y^2} - \frac{y^2}{2} \right) dy + \int_1^2 \left(2 - \frac{y^2}{2} \right) dy + I$$

$$I + \frac{5}{3} - \frac{\pi}{4}$$

$$\text{So, } I = 1 - \frac{\pi}{4} = \int_0^1 \left(1 - \sqrt{1 - y^2} \right) dy$$